

EMPLOYEE SALARY PREDICTION USING MACHINE LEARNING

A Mini Project Report Submitted for Machine Learning Lab

Submitted By

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Abstract

Machine Learning is transforming industries by enabling systems to analyze data and make intelligent predictions. In the corporate sector, predicting employee salaries based on professional qualifications and experience is an important application of Machine Learning.

This project focuses on predicting employee salaries using Machine Learning techniques. The prediction is based on factors such as years of experience, education level, job role, and age. The project uses the Linear Regression algorithm for training and prediction.

The main aim of this project is to demonstrate the practical implementation of Machine Learning concepts in the Human Resource (HR) domain. The system can help companies estimate salaries fairly and assist employees in understanding salary trends.

The project also explains important Machine Learning concepts including data preprocessing, model training, testing, prediction, and evaluation.

1. Introduction

Machine Learning is a branch of Artificial Intelligence that enables systems to learn from data and improve their performance automatically. It is widely used in healthcare, banking, education, finance, and business sectors.

In companies and organizations, salary prediction is useful for recruitment and employee management. Human Resource departments often analyze employee experience, qualifications, and performance before deciding salaries.

Manual salary estimation may lead to inconsistencies and bias. Machine Learning can provide accurate and data-driven salary predictions.

This project implements a Machine Learning model that predicts employee salaries based on professional details such as:

- Years of Experience
- Education Level

- Age
- Performance Score

The project uses the Linear Regression algorithm because salary is a continuous numerical value.

2. Problem Statement

Organizations need an efficient system to estimate employee salaries based on professional qualifications and work experience.

Manual salary prediction can be inaccurate and time-consuming. Therefore, this project aims to build a Machine Learning model that predicts employee salary using employee-related data.

The prediction helps organizations:

- Maintain salary transparency
- Improve recruitment decisions
- Reduce manual effort
- Analyze salary trends

3. Objectives

The objectives of this project are:

1. To understand Machine Learning concepts.
2. To implement Linear Regression practically.
3. To predict employee salary using professional data.
4. To analyze factors affecting salaries.
5. To evaluate model performance.

4. Scope of the Project

The scope of this project includes:

- Salary prediction systems
- HR analytics
- Recruitment support systems
- Employee performance analysis

Future improvements can include:

- Real-world datasets
- Advanced Machine Learning algorithms
- Web-based salary prediction systems
- Real-time employee analytics

5. Literature Review

Machine Learning is widely used in Human Resource analytics for predicting employee turnover, recruitment analysis, and salary estimation.

Researchers have applied algorithms such as:

- Linear Regression
- Decision Tree
- Random Forest
- Support Vector Machine
- Neural Networks

Linear Regression is commonly used for salary prediction because it provides simple and interpretable results.

Studies show that years of experience and educational qualifications strongly influence salary growth.

6. System Requirements

Hardware Requirements

- Processor: Intel i3 or above
- RAM: Minimum 4 GB
- Storage: 500 MB free space

Software Requirements

- Operating System: Windows/Linux
- Python 3.x
- Jupyter Notebook or VS Code

Python Libraries Used

- pandas
- numpy
- matplotlib
- scikit-learn

7. Methodology

The project follows a standard Machine Learning workflow.

Step 1: Data Collection

A dataset containing employee information is created.

The dataset includes:

- Experience
- Age
- Performance Score
- Education Level
- Salary

Step 2: Data Preprocessing

The collected data is cleaned and organized.

Operations performed:

- Handling missing values
- Separating features and target variables
- Formatting data properly

Step 3: Train-Test Split

The dataset is divided into:

- Training Data (80%)
- Testing Data (20%)

Training data is used for model learning, while testing data is used for evaluation.

Step 4: Model Training

The Linear Regression algorithm is trained using employee data.

The algorithm learns the relationship between experience and salary.

Step 5: Prediction

The trained model predicts salaries for test data.

Step 6: Evaluation

The model performance is evaluated using Mean Absolute Error (MAE).

Lower MAE indicates better accuracy.

8. Machine Learning Algorithm

Linear Regression

Linear Regression is a supervised Machine Learning algorithm used for predicting continuous numerical values.

The equation is:

$$Y = mX + c$$

Where:

- Y = Predicted Salary
- X = Input Features
- m = Slope
- c = Constant

Advantages:

- Simple implementation
- Fast computation
- Good for small datasets
- Easy interpretation

9. Dataset Description

The dataset contains employee professional information.

Attribute	Description
Experience	Years of experience
Age	Employee age
PerformanceScore	Performance rating
EducationLevel	Qualification level
Salary	Target variable

Sample Dataset:

Experience	Age	PerformanceScore	EducationLevel	Salary
1	22	60	1	25000
3	25	70	2	35000
5	28	80	2	50000
8	32	90	3	75000

10. Python Implementation

```
# Import libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error

# Create dataset
data = {
    'Experience': [1,2,3,4,5,6,7,8,9,10],
    'Age': [22,23,25,26,28,29,30,32,34,35],
    'PerformanceScore': [60,65,70,72,78,80,85,88,90,95],
    'EducationLevel': [1,1,2,2,2,3,3,3,3,4],
    'Salary': [25000,30000,35000,42000,50000,
               58000,65000,72000,80000,90000]
}

# Create dataframe
df = pd.DataFrame(data)

print(df)

# Features
X = df[['Experience', 'Age',
        'PerformanceScore', 'EducationLevel']]

# Target variable
y = df['Salary']

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)

# Create model
model = LinearRegression()

# Train model
model.fit(X_train, y_train)

# Prediction
predictions = model.predict(X_test)

print("Predicted Salaries:")
```

```

print(predictions)

print("Actual Salaries:")
print(y_test.values)

# Error calculation
mae = mean_absolute_error(y_test, predictions)

print("Mean Absolute Error:", mae)

# Visualization
plt.scatter(y_test, predictions)
plt.xlabel("Actual Salary")
plt.ylabel("Predicted Salary")
plt.title("Actual vs Predicted Salary")
plt.show()

```

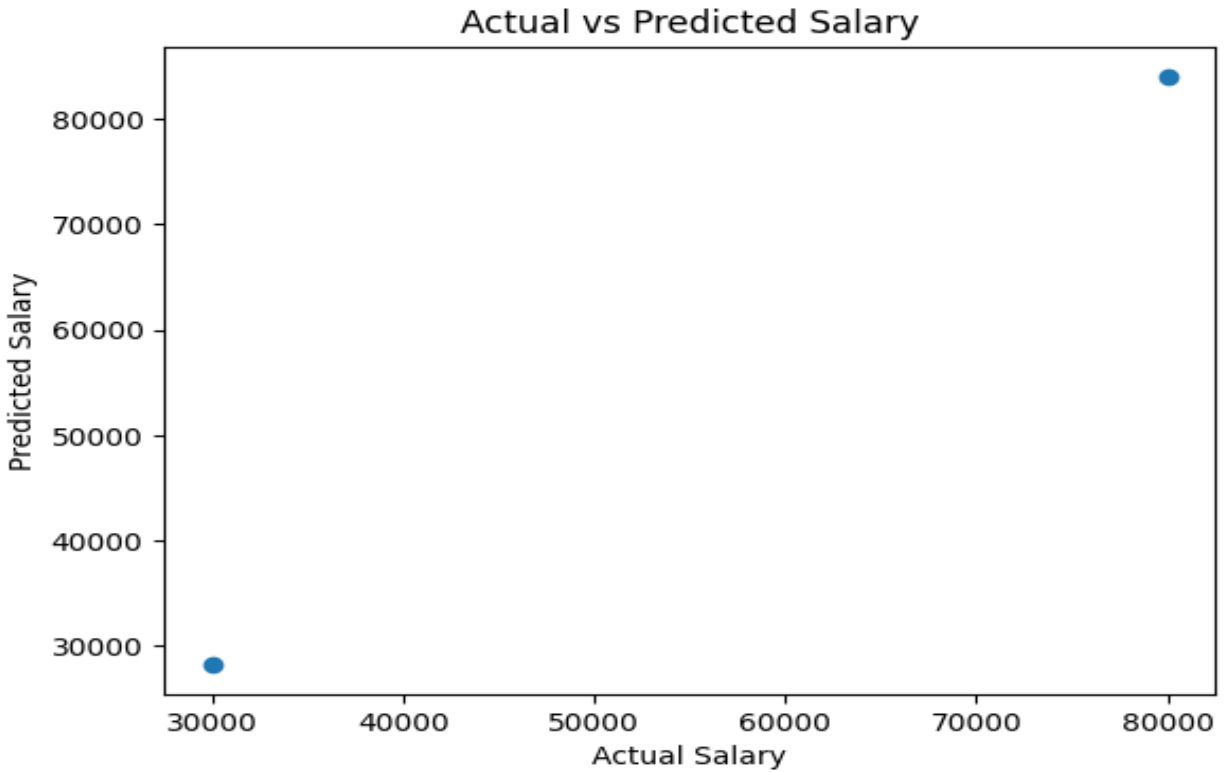
11. Output

Experience	Age	PerformanceScore	EducationLevel	Salary
1	22	60	1	25000
2	23	65	1	30000
3	25	70	2	35000
4	26	72	2	42000
5	28	78	2	50000
6	29	80	3	58000
7	30	85	3	65000
8	32	88	3	72000
9	34	90	3	80000
10	35	95	4	90000

```

Predicted Salaries:
[84019.60784314 28250.          ]
Actual Salaries:
[80000 30000]
Mean Absolute Error: 2884.803921568633

```



12. Advantages

Advantages of the project:

1. Easy implementation
2. Reduces manual salary estimation
3. Helps HR departments
4. Provides quick predictions
5. Demonstrates practical Machine Learning concepts

13. Limitations

Limitations of the project:

1. Uses small dataset
2. Prediction depends on data quality
3. Linear Regression may not capture complex relationships
4. Real-world salaries depend on many additional factors

14. Applications

Applications of Salary Prediction:

- HR analytics
- Recruitment systems
- Salary management systems
- Corporate analytics
- Employee evaluation systems

15. Future Scope

The project can be enhanced by:

1. Using real company datasets
2. Applying Random Forest and XGBoost algorithms
3. Developing a web application
4. Adding data visualization dashboards
5. Implementing employee performance tracking systems

16. Conclusion

This project successfully demonstrates the use of Machine Learning for predicting employee salaries. The Linear Regression algorithm predicts salary based on experience, education level, age, and performance score.

The project highlights the practical application of Machine Learning in the HR domain and shows how data-driven systems can improve decision-making.

17. References

1. Python Documentation – <https://www.python.org/>
2. Scikit-learn Documentation – <https://scikit-learn.org/>
3. Pandas Documentation – <https://pandas.pydata.org/>
4. Machine Learning Concepts by Andrew Ng
5. HR Analytics Research Papers